EOG-Based Robotic Manipulator Control Using SVR Angle Prediction and SVM Direction Classification

Shrawan Rakesh Rane[1], Vinod Santosh Kale[2], Rahul Kumar Yadav[3], Dhanush Balasubramanian[4], Kader shaikh[5]

Automation and Robotics, Vivekanand Education Society's Institute of Technology (VESIT)

Abstract: **Electrooculography (EOG) is a non-invasive technique used to detect eye movement using electrical voltage differences in the region around the eye. This paper presents a control system for controlling a robotic manipulator using EOG. Support Vector Regression (SVR) is used in conjunction with the Support Vector Machine (SVM) algorithm to predict and classify the direction of eye movement. Horizontal and vertical EOG data is captured using the BioAmp EXG Pill, the ADS1115 16-bit Analog to Digital Converter (ADC), and a custom designed Printed Circuit Board (PCB). A well defined target chart is shown to allow accurate association of the EOG data to the correct target location. The dataset is used to train the models. The trained models use the EOG data to predict what angle the user is looking at and determine the direction of the eye movement. This information is then sent to the robotic manipulator using wireless communication with the ESP-NOW protocol. The system uses bio-signals, supervised machine learning, wireless communication, and robotic manipulation to provide an integrated solution. Testing will be conducted to evaluate the system including determining the amount of error in predicted angle, classification performance, communication speed, and accuracy of the robot.**

Keywords: Electrooculography, EOG, Support Vector Regression, Support Vector Machine, Robotic Manipulator, ADS1115, ESP32-C3, ESP-NOW, Human-Robot Interaction.

I. INTRODUCTION

Human-Robot Interaction has become increasingly important within assistive robots, rehabilitation systems and alternate control systems. Traditional robotic manipulators often have control systems involving physical input devices such as joysticks, switches, keyboards or controls. While these are effective they may not suit people who struggle to utilise traditional hand controls. Biosignal input provides an alternate way to control robotic devices by translating biological signals.

Electrooculography works by measuring the potential difference in the eye. The eye can be modelled as an electrical dipole meaning there will be a potential difference between the cornea and retina. As the dipole changes orientation there will be a change in potential seen. Electrodes can be placed close to the eye to measure the horizontal and vertical movement of the eye. This can be used to estimate how the gaze has changed.

Unfortunately, EOG can pose some issues when used to control robotic devices. There can be baseline drift, noise, inter-user variation and electrode variation within the signals. The relationship between the instantaneous EOG signal and the desired gaze angle may not be linear or stationary.

Machine learning algorithms can be used to learn the relationship between the captured EOG signals and the labelled eye movement targets. Support Vector Regression can be used to predict continuous outputs while Support Vector Machine can classify different discrete classes such as left/right/up/down. Together these can be used to predict continuous angles and classify discrete movements.

In this paper an EOG based robotic manipulator system was designed to predict gaze angles using SVR and classify directions using SVM. A structured target chart was used to capture labelled EOG data. This data was then used to train models which were combined with an ESP32 based wireless system using ESP-NOW.

Key Contributions:

- Hardware system developed using BioAmp EXG Pill, ADS1115 16-bit ADC and custom PCB with ESP32-C3 Mini.
- Labelled EOG data captured using structured target chart with multiple reference points.
- Machine learning models developed to predict continuous angles and classify discrete directions.

- Models used to control a robotic manipulator with ESP-NOW.
- System evaluated using signal prediction, classification, communication and robotic control metrics.

II. RELATED WORK

In [1], Duguleana and Mogan created a system to control a robotic system using EOG signals using blink detection. This system had good real time results but focused on discrete commands.

In [2], Abdel-Samei et al. created a wireless robotic arm that could detect horizontal and vertical eye movements. This system used an Arduino, AD620 amplifier and NRF24L01 transceiver. The system would map the EOG signal to specific movement commands.

In [3], Perez Reynoso et al. created a manipulator control system using EOG signals. This used neural networks, genetic algorithms and fuzzy inference. This system also showed how important it was to have a personal calibration.

In [4], Alshehhi et al. compared three different models (SVM, CNN and LSTM) for classifying gestures from EOG. They achieved 82.0%, 90.3% and 93.2% respectively. This showed deep learning can be effective for detecting sequences of eye movements.

In [5], Zendehdel et al. created an EOG system using LSTM to control a UAV system. They achieved 97.6% accuracy for detecting commands. This system had good real time results on workstation and embedded systems.

Research Gap
The research previously has focused on blink detection, gesture classification and neural network control. This work combines Support Vector Regression (SVR) for predicting angles and Support Vector Machine (SVM) for classifying directions. A custom dataset of labelled angles for EOG signals were used. The system uses BioAmp EXG Pill, ADS1115, custom ESP32-C3 Mini PCB and ESP-NOW to control a robotic manipulator wirelessly. The system will be tested for its prediction error, classification accuracy, transmission delay and positioning accuracy.

III. SYSTEM ARCHITECTURE AND METHODOLOGY

A. System Design

The system can be broken down into 5 main sections:

The BioAmp EXG Pill captures the electrical signals generated by eye movement. The ADS1115 converts the analog outputs to a digital measurement. This is then processed by the ESP32-C3 Mini based acquisition unit. The unit contains a custom PCB with the electronics connected.

During training, the user looks at specific target points on the visual chart. The ADC values are captured and labelled using the reference angles and the direction of movement of the target point. This labelled data is used to train and test the SVR and SVM models.

During operation, the EOG data is fed into the trained models. The SVR predicts the desired angle and the SVM classifies the direction of movement. The control data is then sent to the robotic manipulator using ESP-NOW. The controller on the receiving side decodes the command and adjusts the robot movement.

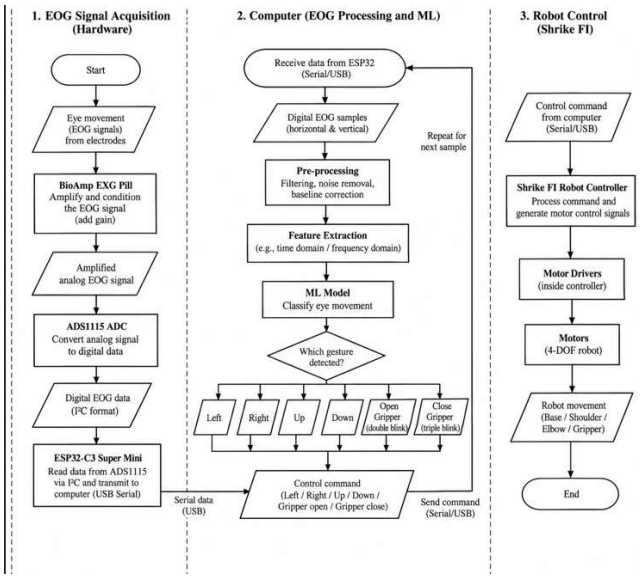


Fig. 1. Proposed EOG-based robotic control system architecture.

B. EOG Signal Acquisition Hardware

The signal acquisition section uses the BioAmp EXG Pill to acquire EOG signals. Electrodes are placed around the eye to measure the voltage differences caused by horizontal and vertical eye movements. The BioAmp signals are then connected to the analog inputs of the ADS1115 ADC.

The ADS1115 is a 16-bit ADC that converts the signals into digital format. The ADC has a higher resolution compared to the default 10-bit ADC on the Arduino Uno.

The horizontal and vertical signals are then converted into digital format. This is done using the Microcontroller based acquisition system. The ESP32-C3 Mini is included in the custom designed PCB.

The hardware includes:

- BioAmp EXG Pill
- ADS1115 16-bit ADC
- ESP32-C3 Mini
- Custom PCB
- Electrodes

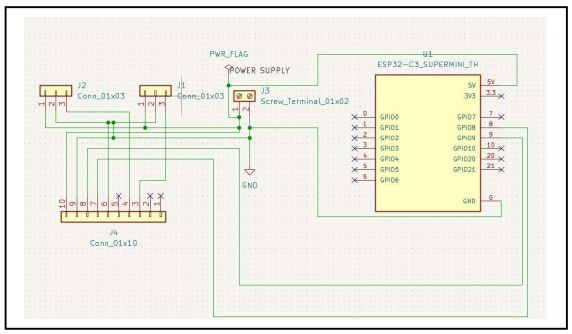


Fig .2. EOG acquisition Schematic

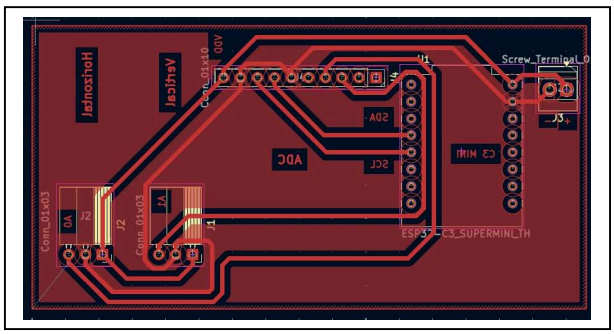


Fig . 3. EOG PCB Layout

The resolution of the ADC will not necessarily give the lowest reliable signal that can be detected. The resolution of the signal will depend on the analog front end, gain, noise, electrode contact, common mode rejection and signal conditioning circuit.

C. Visual Target Chart and Dataset Collection

A visual target chart is used to capture labelled samples. The chart contains multiple circular targets which have been marked. The central target serves as a reference point whilst the target around it can represent different horizontal and vertical gaze directions.

Whilst collecting the dataset the user will be asked to look at a specified target. Once the target has been identified the horizontal and vertical ADC values will be recorded and linked to the known target coordinates. The reference angles for each target can be calculated based on the geometry of the chart and the distance the user is away from the chart.

The dataset will contain captured EOG values and the corresponding target labels. Depending on the training pipeline implemented the label will either contain the reference angle(s) or movement direction.

An example of the dataset structure would be:

- Feature/Label
- Description
- Horizontal EOG
- Digital value from horizontal channel.
- Vertical EOG
- Digital value from vertical channel.
- Horizontal Reference Angle
- Angle of target in horizontal direction.
- Vertical Reference Angle
- Angle of target in vertical direction.
- Direction
- Left, right, up, down or other implemented.
- Sample ID
- Unique identifier for sample/trial.

By collecting the data using the visual target chart it makes the reference targets known allowing supervised learning to be applied. Multiple samples can be collected for each target to get a better understanding of how the signal varies.

The dataset will then be split into training, validation and test data. This can be done at a trial/recording session level. This reduces data leakage issues as samples from the same trial will have high correlation and therefore should not be present in both the training and test data.

D. Reference Angle Calculations

The reference angles for each target location on the visual chart are calculated based on the geometry of the target movement and viewing distance.

The horizontal reference angle is given by:

$$\boldsymbol{\theta_x = \tan^{-1}\left(\frac{x}{D}\right)} \; - \; (1)$$

The Vertical reference angle is given by:

$$\boldsymbol{\theta_x = \tan^{-1}\left(\frac{y}{D}\right)} \; - \; (2)$$

Where:
$\boldsymbol{\theta_x}$ = Horizontal reference angle (degrees)
$\boldsymbol{\theta_y}$ = Vertical reference angle (degrees)
X = Horizontal offset from central reference target
Y = Vertical offset from central reference target
D = Distance between participant's eye and chart

Angles have been converted from radians to degrees using the formula:

$$\boldsymbol{\theta_{\text{degrees}} = \theta_{\text{radians}} \times \frac{180}{\pi}} \; - \; (3)$$

Calculated angles are given as labels to the recorded ADC values. These labelled data sets make up the desired output for training the SVR regression model.

These angles are geometric ideal angles and are different to the predicted EOG angles and robot joint angles.

E. Data Preprocessing and Feature Generation

The EOG data is preprocessed for training and prediction. The preprocessing includes removing any invalid samples and reducing noise from the data, along with removing baseline drift and ensuring the signal is correctly scaled.

The preprocessing will depend on what the pipeline requires. It could involve filtering, baseline removal, normalisation and extracting features from the data. The same preprocessing should be used when training and predicting.

In a two channel system, the input vector may be:

$$\mathbf{x} = [\boldsymbol{E_h, E_v}] \; - \; (1)$$

Where E h and E v are the horizontal and vertical EOG signals/processed features,

If time-based features are to be used then the input vector can also include windowing features, differences in signal or other features extracted from the recorded channels. The exact features used must be the same as those used to train the models.

The data are then provided to the appropriate SVR/SVM training pipeline.

F. SVR-Based Angle Prediction

The Support Vector Regression (SVR) is used to predict continuous values of eye movement angles given the recorded EOG signals. SVR is a regressor which finds a function mapping from the input feature vector to the predicted continuous output.

This can be formulated as a function,

$$\boldsymbol{f}(\mathbf{x}) = \mathbf{w}^{\top}\boldsymbol{\phi}(\mathbf{x}) + \boldsymbol{b} \; - \; (\mathbf{2})$$

Where $\boldsymbol{f}(\mathbf{x})$ represents the input EOG features and $\boldsymbol{\phi}(\mathbf{x})$ Denotes the feature space transformation $\mathbf{w}$ is the Regression coefficient vector, and (b) is the bias term
The SVR estimates a function that minimises the model complexity and the errors in the outputs beyond an (insensitive region. By using a kernel function, non-linear relationships between EOG features and desired angles can be captured.

The trained SVR model takes the extracted EOG features to give a predicted angle:

$$\widehat{\boldsymbol{\theta}} = \boldsymbol{f}_{\mathbf{SVR}}(\mathbf{x}) \; - \; (3)$$

If we are predicting both the horizontal and vertical angles separately, then we could train two individual regression models. This should be specified in the final implementation.

The predicted angle is then converted into an appropriate control signal for the robot, based on the robots allowable movement and implemented safety constraints.

G. SVM-Based Direction Classification

A Support Vector Machine classification is performed to determine the direction of the collected EOG data. This is achieved by performing a supervised classification problem using labelled examples within the dataset.

The classifier learns decision boundaries to separate between the different movement classes. These classes will depend on the dataset implemented and may consist of left, right, up and down movements.

The prediction can be given as

$$\widehat{\boldsymbol{d}} = \boldsymbol{f}_{\mathbf{SVM}}(\mathbf{x})$$

Where **f(x)** is the input EOG features and $\hat{\mathbf{d}}$ is the predicted movement class.

The SVM output gives a discrete directional command whilst the SVR gives a continuous angular estimate.

The classification output can be used to determine the desired direction to move in whilst the regression output determines what angle to move to. The exact command generation should reflect how you will implement the command, this may include determining which axis to move, what to do in a neutral state or implementing any movement constraints.

H. UART-Based Serial Communication

The UART communication protocol has been implemented to enable data transfer between the EOG acquisition hardware, processing unit and robot controller.

The ESP32-C3 Mini device collects the digitalised horizontal and vertical EOG data using the ADS1115 ADC. This data is then sent to the processing unit using the UART serial communication protocol. The data is then processed and passed into the trained SVR and SVM models.

The processing unit takes the outputs from the model and creates the appropriate commands for the robot. These commands are sent using the second UART serial communication protocol to the Shrike-FI ESP32-S3 controller. The Shrike-FI then receives these commands and processes them. The appropriate commands are then sent to the servos of the robotic manipulator.

This allows for a serial connection to be setup between the EOG acquisition, prediction using machine learning and robotic control. The UART protocol enables a direct connection between the hardware components.

The communication subsystem has been assessed based on its reliability, delay using serial communication, accuracy of commands and responsiveness of the robotic manipulator.

I. Robotic Manipulator Control

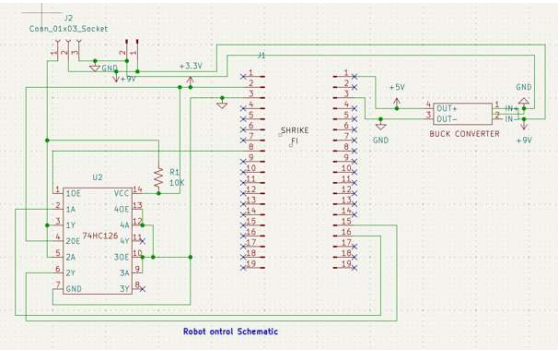


Fig .4. Robot Control Schematic

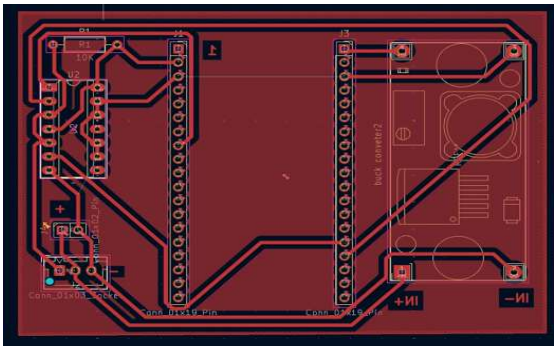


Fig .5. Robot Control PCB layout

The robotic manipulator is a physical output of the system. The Shrike-FI ESP32-S3 controller takes commands from the processing system via UART serial communication. It then determines the robot's movement.

The predicted angle from the SVR model is converted to a joint target/control variable. The direction information from the SVM classifier is used to determine what movement is intended. The commands are processed by the Shrike-FI controller, and passed to the HTS-35H bus servos. Limits are placed to ensure commands do not exceed the robots allowable range.

This creates a system that can take eye movement data, convert it into a prediction using machine learning, send it via UART serial communication and cause the robot to move.
When evaluating the system, it is important to note that the predicted angle from the SVR model, the target position and actual joint position may not be the same.

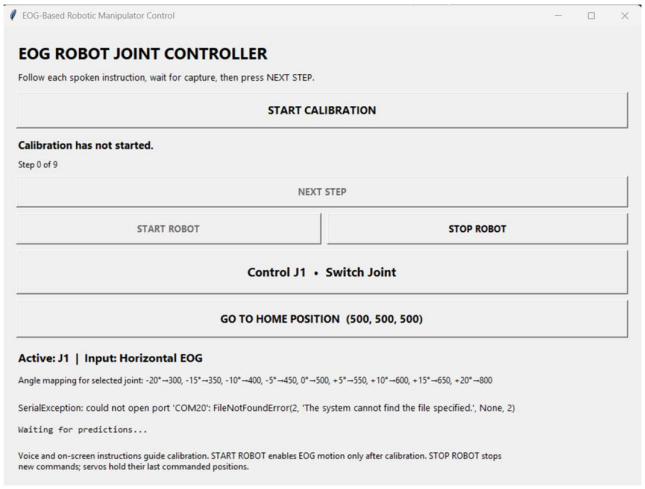


Fig.6. Robot Control GUI

IV. EXPERIMENTAL SETUP AND PERFORMANCE EVALUATION

A. Experimental Setup

The BioAmp EXG Pill, ADS1115 ADC and custom ESP32-C3 Mini PCB were used along with a visual target chart, trained ML models, UART serial connections and robotic manipulator.

The participant would view the targets on the chart whilst the EOG hardware captured the data. The samples were labelled with the target they corresponded to.

The trained models were then implemented into the control system. During the test phase, the ESP32-C3 Mini would capture new EOG data and send it to the system via UART serial connection. The SVR and SVM models would process the data to predict the angle of eye movement and classify the direction. The models would then produce control commands that would be sent via another UART serial connection to the Shrike-FI ESP32-S3 controller. The robotic manipulator would then be controlled to produce a response. This response would be recorded to test the overall system.

Experimental Parameters:

Number of participants and sessions: 1 participant 23(sessions).
Number of targets and number of trials per target: 9 horizontal and vertical targets (-20°, -15°, -10°, -5°, 0°, +5°, +10°, +15°, +20°).
Sampling rate:250Hz. ADS1115 used as 16-bit ADC. Gain set to GAIN_ONE = ±4.096V. Resolution of approx 125μV per bit.
Preprocessing and Feature Design:

- 50Hz notch filter. Q = 30.
- Bandpass filter from 0.1-10Hz.
- Kalman filter. Q = 0.001 and R = 2.0.

Window size: 125 samples (0.5s).
Overlap: 50%. Hop size: 62 samples.
Horizontal and vertical EOG data used as inputs.
SVR: R^2 = 0.85, **RMSE =** 38.73, **MAE =** 30.90.
SVM: Accuracy = 0.70, **Balanced Accuracy** = 0.65, **Macro F1** = 0.62.
Training, validation and test split:70(training) 30 (testing).
UART:115200 baud rate, 8 data bits, no parity, 1 stop bit (8N1). Two serial communications. ESP32-C3 Mini to PC, PC to **Shrike-FI ESP32-S3. Robot joint limits and reference target angle calculation:** Joint 1 command range is 400-800 which corresponds to reference angles of -20° to +20°, in 5° increments. Joints 2 and 3 are fixed to 500. Reference target angles are calculated using geometry: 45cm from points and 5degree angles apart from each other.
Joint position: Shrike-FI ESP32-S3 sends target position commands to the HTS-35H bus servos. Position data can be obtained from the bus.

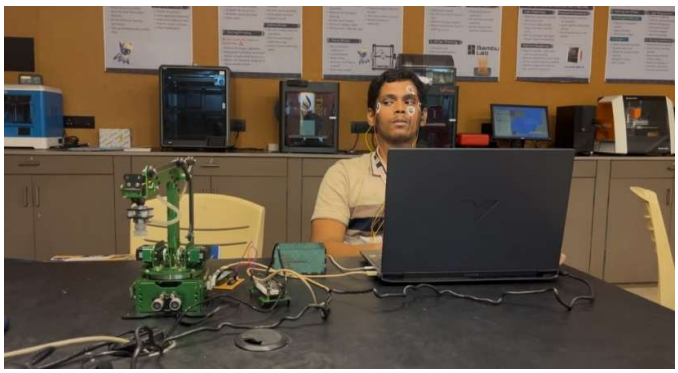


Fig.7. Experimental setup of the EOG-based robotic manipulator control system.

B. Angle Prediction Evaluation

The performance of the SVR model can be evaluated by comparing the actual angles to those of the target chart.

Mean absolute error (MAE) is calculated as

$$\mathbf{MAE} = \frac{\mathbf{1}}{\boldsymbol{N}}\sum_{\boldsymbol{i=1}}^{\boldsymbol{N}}\left|\boldsymbol{\theta_i} - \widehat{\boldsymbol{\theta}}_{\boldsymbol{i}}\right| \text{ - (1)}$$

Root mean square error (RMSE) is calculated as

$$\mathbf{RMSE} = \sqrt{\frac{\mathbf{1}}{\boldsymbol{N}}\sum_{\boldsymbol{i=1}}^{\boldsymbol{N}}\left(\boldsymbol{\theta_i} - \widehat{\boldsymbol{\theta}}_{\boldsymbol{i}}\right)^{\mathbf{2}}} \text{ -(2)}$$

where N is the number of samples in the test data set, θ is the true angle, and $\hat{\theta}$ is the predicted angle.
The MAE tells us the average error we can expect from our model. The RMSE weights the larger errors more heavily. Both should be reported in degrees if we are regressing in degrees.

If predicting horizontal and vertical angles separately then these values should be calculated and reported separately.

C. Direction Classification Evaluation

The performance of the SVM classifier can be evaluated using the labelled test data set. The classification accuracy can be found using,

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \times 100$$

Accuracy can also be calculated for each direction class using precision, recall and F1 score. A confusion matrix can also be used to find any systematic errors. For example, movements to the left may have been classified as a right movement or vertical movements may have been confused.

Evaluation should be performed using samples not used during training or tuning.

D. End-to-End Control Evaluation

Alongside evaluating individual model metrics, it is recommended to also evaluate the entire control pipeline. The end-to-end latency of each part of the pipeline should be measured, including: signal acquisition, pre-processing, model prediction, UART serial communication between the ESP32-C3 Mini and PC, command generation, UART serial communication between the PC and Shrike-FI ESP32-S3, processing at the receiver and robot start-up time.

If possible report the delay caused by communication versus physical movement. This can help to determine which part of the system is causing delays in terms of signal processing, UART communication and robot actuation.

Recommended metrics to record:

- Mean Absolute Error (MAE) of angle prediction using the SVR.
- Accuracy of classifier using the SVM. F1-score per class.
- UART Communication reliability. Latency of serial communication.
- End-to-end command to response latency.
- Error of robot reaching desired position. Rate of successfully executed commands.
- Reproducibility.

E. Results and Discussion

Results of the experiments should be shown using the data collected based on the trained models and robot testing. Complete the tables below:

[1] TABLE I. SVR ANGLE PREDICTION PERFORMANCE

Metric	Horizontal axis
MAE (degrees)	30.90°
RMSE (degrees)	38.73°
R^2 score	0.85
Predictions within ±5°	25.03%

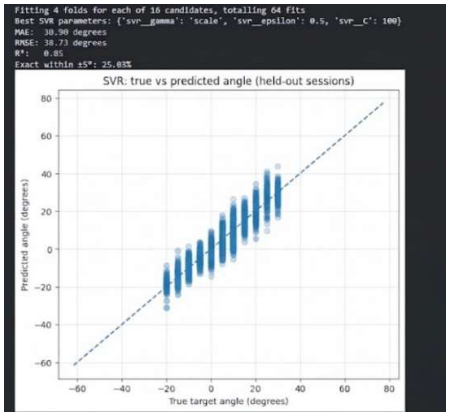


Fig.8. SVR-predicted versus actual target angles on held-out test sessions.

TABLE II. SVM DIRECTION CLASSIFICATION PERFORMANCE

```
Fitting 4 folds for each of 12 candidates, totalling 48 fits
Best SVM parameters: {'svc__gamma': 'scale', 'svc__C': 100}
Accuracy:          0.7000
Balanced accuracy: 0.6500
Macro F1:          0.6200

Classification report:

              precision    recall  f1-score   support

     BLINK_1       0.61      0.58      0.59       150
     BLINK_2       0.65      0.62      0.63       150
     BLINK_3       0.63      0.60      0.61       150
        DOWN       0.68      0.72      0.70       850
        LEFT       0.73      0.69      0.71      2200
        REST       0.55      0.50      0.52       100
       RIGHT       0.78      0.75      0.76      3600
          UP       0.71      0.67      0.69      1300

    accuracy                           0.70      8500
   macro avg       0.66      0.64      0.62      8500
weighted avg       0.71      0.70      0.70      8500
```

Fig. 9. SVM direction classification performance and classification report.

Metric	Measured result
Test accuracy (%)	70.00%
Balanced accuracy (%)	65.00%
Macro-averaged precision	0.66
Macro-averaged recall	0.64
Macro-averaged F1-score	0.62

TABLE III. UART COMMUNICATION AND ROBOTIC CONTROL PERFORMANCE

Metric	Result
UART communication reliability (%)	100% – No data loss observed during testing
Average UART communication latency (ms)	0.087 ms
Average end-to-end response time (ms)	Approximately 2–5 seconds
Mean robot target-position error (degrees)	+- 5%
Successful command execution rate (%)	100% observed – All transmitted commands were received and robot movement was demonstrated

V. LIMITATIONS AND FUTURE WORK

There are several factors that may affect the performance of the proposed system. These include variation in amplitude of EOG signals, positioning of electrodes, baseline drift, muscle movement and individual differences in eye movement. If a small amount of data was used to train the model it may not generalise well for different users and electrode setups.

The target angle depends on how accurate the chart and distance are. The relation between the captured EOG signals and gaze angle may also change over time meaning calibration must be performed.

Further work could look into improving signal conditioning techniques, automatically correcting baseline drift, improving feature extraction, hyperparameter tuning and evaluating the model independent of users. More research can be done looking at how the system performs across multiple people and sessions.

The wireless control component could be further improved using monitoring techniques, command validation, timeout functionality and a safe stop. Evaluation using more data and multiple physical experiments can be performed to ensure the full EOG control system is reliable.

VI. CONCLUSION

This paper describes a brain-computer interface using EOG signals to control a robotic manipulator. It features a Support Vector Regression based continuous angle prediction system, Support Vector Machine based direction prediction system and ESP-NOW wireless communication protocol. The system includes a BioAmp EXG Pill for capturing EOG signals, an ADS1115 16-bit ADC and a custom PCB with an ESP32-C3 Mini.

A visual target display is used to provide the correct reference positions when capturing data from the ADC. This results in a labelled dataset that can be used for the supervised learning models. These can then be used for predicting the desired angle of eye-movement and classifying if the eye is moving in a particular direction. These predictions are then sent wirelessly to a controller which will issue commands to the robotic manipulator.

This design combines many aspects including physiological signal capture, machine learning, microcontroller design, wireless communication and robotic components. The system needs testing to see how effective it is. This can be done by analysing the

error from the regression model, classification metrics and communication speed. This will give an indication of how reliable and usable the system design is.

ACKNOWLEDGMENT

The authors would like to thank Prof. Sangeeta Prasanna (Head, Department of Automation and Robotics, Vivekanand Education Society's Institute of Technology, Mumbai), for guiding and encouraging us throughout the completion of our project.

We thank our project guide, Prof. Kader Shaikh for his guidance on time-to-time, valuable suggestions and support for doing this project of controlling robotic manipulator using Electrooculogram.

We would also like to thank Prof. N. Gopalakrishnan for his valuable guidance throughout our project.
Finally we thank the Department of Automation and Robotics, Vivekanand Education Society's Institute of Technology, for providing an academic environment which helped in completing this research work.

REFERENCES

[1] C. Cortes and V. Vapnik, "Support-vector networks," Machine Learning, vol. 20, pp. 273-297, 1995.

[2] A. J. Smola and B. Schölkopf, "A tutorial on support vector regression," Statistics and Computing, vol. 14, pp. 199-222, 2004.

[3] C.M. Bishop, Pattern Recognition and Machine Learning, New York, NY, USA: Springer, 2006.

[4] Texas Instruments, ADS1113, ADS1114, ADS1115: Ultra-Small, Low-Power, I2C-Compatible, 860-SPS, 16-Bit ADCs With Internal Reference, Oscillator, and Programmable Comparator, [Online]. Available: https://www.ti.com/lit/ds/symlink/ads1113.pdf. [Accessed: 25 May 2022].

[5] Espressif Systems, ESP-NOW, ESP-IDF Programming Guide. [Online]. Available: https://esp-idf.readthedocs.io/en/latest/api-reference/system/espnow.html. [Accessed: 25 May 2022].

[6] Upside Down Labs, BioAmp EXG Pill. [Online]. Available: https://www.upside-downlabs.com/bioamp-exg-pill/. [Accessed: 25 May 2022].

[7] Research articles verifying the use of EOG for tracking eye movements, estimating gaze angles and controlling robotic manipulators using biosignals.

[8] Esp32-C3 Mini development board/module documentation.